

OpenAI Math Digest

CATALOGUE 063

The generalized Mukai conjecture

Catalogue guide & one proof overview

Independent reading guide · English edition

Edition: 2026-10-09

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Contents

The catalogue guide is followed by the article. The contents and citations are clickable. Wide tables are arranged as entries for print; the claims and source references are retained.

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Original-manuscript snapshot: fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb

Catalogue 063 · Guide

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CATALOGUE 063

The generalized Mukai conjecture

This catalogue concerns the generalized Mukai inequality for smooth complex Fano manifolds and the classification at equality. The official list contains one manuscript.

All manuscripts, in official catalogue order

These summaries give the principal scope; consult the cited original results for the complete hypotheses and statements.

Catalogue 063 · 2026-10-09 checked source snapshot

#

01

Manuscript and overview

The generalized Mukai conjecture →

Short name: Generalized Mukai

2026-09-24 · 19 pages

[Original paper \(GitHub\) ↗](#)

Principal claim

For smooth complex projective Fano varieties, asserts $\rho(X)(\iota(X)-1) \leq \dim X$ and characterizes equality by products of projective spaces.

[Theorem 1.1 · p. 2 ↗](#)

Role and available overview

A descendant lower bound and a recurrence give independent eigenvalues; an alternating trace gives the inequality. Rational-curve families and a product characterization settle equality.

[Read the proof overview →](#)

Connections to external results

The [input table](#) distinguishes direct inputs from an alternative contraction, with uses, hypotheses, and checking scope. Connections to other catalogues remain incompletely investigated.

Sources and checking scope

Official subject: **Algebraic and complex geometry**. Catalogue edition: October 6, 2026.

[OpenAI Research Catalog](#) · [Catalogue 063](#) · [p. 8](#) · [Official manuscript list](#)

Source snapshot: `fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb`.

Sections 2–6 and the statements and applications of direct inputs were checked. This is not a complete independent proof verification; remaining checks are listed [in the article](#).

063 / 01

[Read on the website ↗](#)

CATALOGUE 063 · PAPER 01 / 1 Overview

The generalized Mukai conjecture

Point-descendant lower bounds, independent quantum divisors, and rigidity through minimal rational curves

The manuscript claims the Picard-number–pseudoindex inequality for smooth complex Fano manifolds by exploiting the grading of quantum multiplication. Its central task is to construct a nonzero product of correspondences in independent curve degrees. Only at equality does it pass to actual families of minimal rational curves and invoke an existing characterization of products.

Manuscript author: OpenAI · 2026-09-24 · 19 pages, including references

[Read the manuscript on GitHub ↗](#)

1. Main results

Theorem 1.1 — The generalized Mukai inequality and equality

Let X be a smooth connected complex projective Fano manifold of positive dimension n , Picard number ρ_X , and pseudoindex ι_X . The manuscript asserts

$$\rho_X(\iota_X - 1) \leq n$$

with equality if and only if

$$X \simeq (\mathbb{P}^{\iota_X-1})^{\rho_X}.$$

Write $r = \rho_X$ and $\iota = \iota_X$. If $\iota = 1$, the left side is $0 < n$, so the argument concerns $\iota \geq 2$.

[Theorem 1.1 · p. 2; §1 · pp. 2–3](#)

References on the arrows

Unprefixed theorem, lemma, section and equation numbers refer to [Generalized Mukai](#). External inputs use the following keys.

- [KM](#) Kontsevich–Manin — Correlator relations: Correlator relations and recursion. 1998 published version.
- [PA](#) Rahul Pandharipande — Rational curves on hypersurfaces: Divisor, string and topological recursion relations. 1998 published version (Astérisque 252).
- [MM](#) Andrei Mustață and Magdalena Anca Mustață — Intermediate moduli: An alternative contraction via intermediate moduli of stable maps. arXiv v4 (July 17, 2006).
- [MMC](#) Anca M. Mustață and Andrei Mustață — Chow ring: Intermediate stacks and the cotangent class. arXiv v5 (November 30, 2006).
- [TZ](#) Zhiyu Tian and Hong R. Zong — One-cycles: Generation of one-cycles by rational curves and comb smoothing. arXiv v5 (July 22, 2013).
- [CA](#) Frédéric Campana — Fano rational connectedness: Rational connectedness of Fano varieties. 1992 published version.
- [DE](#) Olivier Debarre — Rational curves: Passage to separable rational connectedness. Lecture notes dated August 29, 2011.
- [BCDD](#) Laurent Bonavero, Cinzia Casagrande, Olivier Debarre and Stéphane Druel — Mukai chain bound: Dimension bound for an ordered chain locus. 2003 published version.
- [ERR](#) Bonavero–Casagrande–Debarre–Drueel — Mukai chain bound: Correction to the hypotheses of the chain bound. Undated one-page author correction (retrieved October 9, 2026).
- [OC](#) Gianluca Occhetta — Product characterization: Characterization of products of projective spaces. 2006 published version.

Each citation gives the result and pages. When printed pagination differs from the PDF position, both are shown. GitHub previews do not automatically jump to the cited page.

2. Theorem 1.1 — Proof overview of the main theorem

Proof target : [Theorem 1.1](#) · p. 2; §1 · pp. 2–3

Proof route. Use X, n, r, ι from Theorem 1.1. If $\iota = 1$, then $r(\iota - 1) = 0 < n$, so the inequality is immediate and equality is impossible. Assume $\iota \geq 2$ below. The lower bound in [Proposition 3.1](#) leads to independence in [Proposition 4.1](#). [Proposition 5.3](#) supplies both the inequality and the input for the equality case; [Proposition 6.4](#) then gives the product structure.

1 A quantitative descendant bound in free degrees

Proposition 3.1: a lower bound incompatible with superexponential decay
 $\langle \tau_{d-2}(\text{pt}) \rangle_\beta \geq (ad)^{-(d-1)}$

Add the recurrence and density of free degrees; dependence gives a contradiction.

Lemma 2.1; (4)–(5) · p. 5

Lemma 3.3 · pp. 10–11

Proposition 4.1 (proof) · p. 12

Proposition 4.1; (8)–(11) · pp. 11–13

2 An independent joint tuple of quantum divisors

Proposition 4.1: $\lambda_1, \dots, \lambda_r$ are algebraically independent over $\mathbb{C}$.

Extract an independent nonzero term from the alternating trace; count the grading drop.

Lemmas 5.1–5.2 · pp. 13–15

Proposition 5.3; (15) · p. 15

3 An independent nonzero product and the inequality

$S_{\gamma_1} \cdots S_{\gamma_r} \neq 0, \quad \gamma_1, \dots, \gamma_r \text{ independent}$

$$r(\iota - 1) \leq \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n$$

Equality makes every degree ι . Pass to actual curve families and make all of them covering.

Lemma 6.3 · pp. 16–17

Theorem 6.1 · p. 16

Proposition 6.4 · p. 17

Theorem 6.2 · p. 16

Proposition 6.4; converse · p. 17

4 Classification at equality

$$r(\iota - 1) = n \implies X \simeq (\mathbb{P}^{\iota-1})^r$$

Proposition 6.4 uses the independent nonzero product obtained above.

Check the pseudoindeces using lines in the factors and the anticanonical divisor.

Theorem 6.2 · p. 16

Proposition 6.4; converse · p. 17

5 Compute the product invariants for the converse

$$\dim X = r(\iota - 1), \quad \rho_X = r, \quad \iota_X = \iota$$

For $\iota = 1$, start with $0 < n$. All conclusions of Theorem 1.1 now follow.

Theorem 1.1: from the descendant bound to the inequality, equality classification, and converse

[Open figure ↗](#) [TeX source ↓](#)

1. From a lower bound and independence to the inequality

Proposition 3.1 alone does not give independence. Combine it with the recurrence in Lemma 2.1 and the density of free degrees in Lemma 3.3. If every joint eigenvalue tuple were algebraically dependent, the resulting superexponential upper bound would contradict the descendant lower bound. This proves Proposition 4.1. The details are separated into the sections on the [lower bound](#) and [independence](#).

Proposition 3.1 · p. 6 · Lemma 2.1; (4)–(5) · p. 5 · Lemma 3.3 · pp. 10–11; Proposition 4.1 (proof) · p. 12 · Proposition 4.1; (8)–(11) · pp. 11–13

Apply Proposition 5.3 to the independent tuple. It yields linearly independent curve degrees $\gamma_1, \dots, \gamma_r$ with $S_{\gamma_1} \cdots S_{\gamma_r} \neq 0$. Writing $d_\gamma = -K_X \cdot \gamma$, one obtains

$$r(\iota - 1) \leq \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n.$$

The first inequality uses $d_{\gamma_j} \geq \iota$ for positive stable-map degrees; the second counts the grading drop of the nonzero product. This proves the inequality in the main theorem. The [alternating-trace section](#) explains how to extract this product.

Proposition 5.3; (15) · p. 15

2. Equality forces the product of projective spaces

If $r(\iota - 1) = n$, all the preceding bounds are equalities and each $d_{\gamma_j} = \iota$. The independent nonzero product from Proposition 5.3 is exactly the additional input required by Proposition 6.4. Lemma 6.3 passes to actual proper rational-curve families. The corrected chain-locus dimension bound and cyclic rotation make every family covering. The product characterization gives

$$X \simeq (\mathbb{P}^{\iota-1})^r.$$

The equality section explains the hypotheses and applications of these external results.

Lemma 6.3 · pp. 16–17 · Proposition 6.4 · p. 17 ·
Mukai chain bound [BCDD] · Theorem 5.2 · print p. 623 / PDF p. 23 ·
Mukai chain bound [ERR] · Erratum · p. 1 ·
Product characterization [OC] · Theorem 1.1 · print p. 271 / PDF p. 2

3. The product attains equality

Conversely, let $X = (\mathbb{P}^{\iota-1})^r$ with $\iota \geq 2$. Its dimension is $r(\iota - 1)$ and its Picard number is r . If H_j is the pullback of the hyperplane class from the j th factor, then $-K_X = \iota \sum_j H_j$. A line in one factor has anticanonical degree ι . Every irreducible rational curve has nonnegative integral intersections with the H_j , at least one of which is positive. Thus the pseudoindex is ι , and equality holds. This completes both directions of Theorem 1.1.

Theorem 6.2 · p. 16; Proposition 6.4; converse · p. 17

3. Proof of Proposition 3.1 — the point-descendant lower bound

Proof target : Proposition 3.1 · p. 6

Proposition 3.1 — Point-descendant lower bound

Let X be the smooth complex projective Fano manifold above, with $\iota \geq 2$. Fix a positive integer a and an embedding $X \hookrightarrow \mathbb{P}^N$ such that $\mathcal{O}_{\mathbb{P}^N}(1)|_X \simeq \mathcal{O}_X(-aK_X)$. Suppose that the numerical curve degree β is represented by a nonconstant free map $\mathbb{P}^1 \rightarrow X$. Put $d = -K_X \cdot \beta$ and $b = ad$. The manuscript asserts

$$\langle \tau_{d-2}(\mathbf{pt}) \rangle_\beta \geq b^{-(d-1)}.$$

Proposition 3.1 · p. 6

Write $d_\beta = -K_X \cdot \beta$. The next section compares this quantitative bound with an upper bound from the recurrence.

1 Evaluation fiber and unobstructed locus

$F = \text{ev}^{-1}(x)$, $h = d - 2$, $b = ad$, $[F]^{\text{vir}}|_U = [U]$
 β is free; x is very general; U has irreducible source.

Move all degree to the marked component.

§§3.2–3.3 · pp. 7–8

Alternative contraction preserving the marking:

Intermediate moduli [MM] · Proposition 1.3 · pp. 5–6

2 Contract unmarked trees to common zeros

$\Phi : F \rightarrow W$, $\psi = \Phi^*\xi$, $1 \leq w_j \leq b$

The marking and cotangent line are preserved; generic degree on U is one.

Use degree $\geq \iota \geq 2$ for each tail and Chow localization.

§3.4 · pp. 8–9

Effectivity of the virtual class itself is not required.

3 Kill the pushed-forward boundary

$\dim \Phi(F \setminus U) \leq d_0 + u - 2 \leq h - 1$

$\Phi_*[F]^{\text{vir}} = \sum_{\alpha} [\Phi(U_{\alpha})] \neq 0$ effective h -cycle

Use coordinate-substack degree $1/(w_0 \cdots w_h)$ and projection formula.

Lemma 3.2; Proposition 3.1 (proof) · pp. 9–10

4 A quantitative bound from weighted degree

$$\langle \tau_{d-2}(\text{pt}) \rangle_{\beta} = \int_{\Phi_*[F]^{\text{vir}}} \xi^h \geq b^{-(h+1)} = (ad)^{-(d-1)}$$

Contracting the evaluation fiber and bounding the boundary image to obtain the descendant lower bound

Open figure ↗ TeX source ↓

1. A contraction preserving the unobstructed open part

Let β be free, and set $d = d_{\beta}$, $b = ad$, and $h = d - 2$. For a very general point x , put $F = \text{ev}^{-1}(x) \subset \overline{M}_{0,1}(X, \beta)$. The open substack U with irreducible source is nonempty and smooth, with $[F]^{\text{vir}}|_U = [U]$. The manuscript retains the marked component and replaces the other trees by common zeros at their attachment points. In §3.2, $L^+ = L(\sum_e e\Delta_e)$ transfers all degree to the marked component without changing a neighborhood of the marking. Normalized coefficients define $\Phi : F \rightarrow W$ into a weighted projective stack W , with $\psi = \Phi^*\xi$, $\xi = c_1(\mathcal{O}_W(1))$, and generic stack degree one on U .

§§3.1–3.3 · pp. 6–8 ·

Intermediate moduli [MM] · Definition 1.2; Proposition 1.3 · pp. 5–6; Lemma 3.3 (proof) · pp. 17–18 ·

Chow ring [MMC] · Proposition 1.7 · p. 5

2. It is the boundary pushforward that vanishes

Let d_0 be the anticanonical degree of the marked component and u the number of attached trees. The retained data have dimension at most $d_0 + u - 2$. Each tree has degree at least $\iota \geq 2$, so $d - d_0 \geq 2u$ yields

$$\dim \Phi(F \setminus U) \leq d_0 + u - 2 \leq h - 1.$$

The boundary term in Chow localization therefore vanishes after pushforward. Thus $\Phi_*[F]^{\text{vir}}$ is a nonempty effective h -cycle consisting of closures of components of $\Phi(U)$. The argument does not assume that $[F]^{\text{vir}}$ itself is effective or that all tail maps are unobstructed.

§3.4 · pp. 8–9

3. Turning stack degrees into a numerical bound

All weights of W are at most b . Lemma 3.2 gives $\int_Z \xi^h \geq b^{-(h+1)}$ for every integral h -dimensional closed substack $Z \subset W$. A monomial degeneration of the weighted homogeneous ideal reduces the estimate to coordinate substacks, of degree $1/(w_0 \cdots w_h)$. The projection formula gives $\int_{[F]^{\text{vir}}} \psi^h = \int_{\Phi_*[F]^{\text{vir}}} \xi^h$, hence the desired lower bound. Stabilizer denominators are essential to this numerical estimate.

Lemma 3.2; Proposition 3.1 (proof) · pp. 9–10

4. Proof of Proposition 4.1 — independent eigenvalues

Proof target : Proposition 4.1; (8)–(11) · pp. 11–13

Put $H_k = H^{2k}(X, \mathbb{C})$ and $H = \bigoplus_{k=0}^n H_k$. For a numerical divisor basis $D_1, \dots, D_r$, set $\beta_j = D_j \cdot \beta$. Define the two-point correspondence by $(S_\beta a, b) = \langle a, b \rangle_\beta$. The commuting quantum multiplication matrices are

$$A_j(q) = D_j \cup (-) + \sum_{\beta > 0} q^\beta \beta_j S_\beta, \quad S_\beta(H_k) \subset H_{k+1-d_\beta}.$$

The grading kills terms with $d_\beta > n + 1$, so these are finite Laurent polynomial matrices. No semisimplicity of quantum cohomology is assumed.

Proposition 4.1 — An algebraically independent joint tuple

For X and the quantum divisor operators $A_1(q), \dots, A_r(q)$ above, take a simultaneous upper triangular form over $L = \overline{\mathbb{C}(q_1, \dots, q_r)}$. A joint diagonal tuple consists of the entries at one common diagonal position. The manuscript asserts that at least one such tuple $(\lambda_1, \dots, \lambda_r)$ has coordinates algebraically independent over $\mathbb{C}$. Semisimplicity of quantum cohomology is not assumed.

Proposition 4.1 · p. 11

1 Factorially normalize the descendants

$$\beta_j v_\beta = \sum_\gamma A_{j,\gamma} v_{\beta-\gamma}, \quad w_{\beta,k} = (d_\beta + k)! v_{\beta,k}$$

$$\|w_\beta\| \leq B^{d_\beta+1}, \quad (v_\beta, 1) = \langle \tau_{d_\beta-2}(\text{pt}) \rangle_\beta$$

Recurrence and factorial normalization give an exponential bound.

[Lemma 2.1 · p. 5](#)

[§4.1 · pp. 11–12](#)

[Correlator relations \[KM\] · Lemma 1.4 · print p. 391 / PDF p. 7](#)

[Rational curves on hypersurfaces \[PA\] · §1.2 · print p. 311 / PDF p. 6](#)

[Homogeneity and density of free degrees:](#)

[Lemma 3.3; §4.2 · pp. 10–12](#)

2 Turn dependence into a homogeneous relation

All joint tuples dependent $\implies Q(A_1, \dots, A_r) = 0, Q \neq 0$

Choose a free degree η with $Q(\eta) \neq 0$.

Use generation, comb smoothing, and finite shifts.

[One-cycles \[TZ\] · Theorem 1.3 · p. 2](#)

[Proposition 2.4 · p. 4](#)

[\(9\)–\(11\) · pp. 12–13](#)

3 Iterate finite shifts linearly many times

$$\|w_{s\eta}\| \leq (2C/d)^{\lfloor \epsilon d \rfloor} B^{2d+1}, \quad d = d_{s\eta}$$

Each step gains $O(1/d)$; degree and direction stay controlled.

Compare logarithms $-\epsilon d \log d + O(d)$ and $-O(d)$.

[Proposition 3.1 · p. 6](#)

[Proposition 4.1 · p. 11](#)

4 An exponential lower bound forces independence

$$(w_{s\eta}, 1) \geq ad^{n+1}(ae)^{-d}$$

Some joint tuple $\lambda_1, \dots, \lambda_r$ is algebraically independent over $\mathbb{C}$.

Contradicting superexponential decay with the exponential lower bound for normalized descendants

[Open figure ↗](#) [TeX source ↓](#)

1. A recurrence with a separate degree-zero initial value

Set $v_0 = \text{pt}$ and, in positive degree, $(v_\beta, b) = \sum_{\ell \geq 0} \langle \tau_\ell(\text{pt}), b \rangle_\beta$. Lemma 2.1 gives

$$\beta_j v_\beta = \sum_\gamma A_{j,\gamma} v_{\beta-\gamma}, \quad (v_\beta, 1) = \langle \tau_{d_\beta-2}(\text{pt}) \rangle_\beta.$$

The divisor correction vanishes because $D_j \cup \mathbf{pt} = \mathbf{0}$. In topological recursion, the degree-zero three-point side contributes classical cup product, while the primary term contributes $A_{j,\beta}v_0$. The initial value v_0 must not be identified with an unstable degree-zero two-point invariant. The second identity uses the string equation. The cited source explicitly states that this equation remains valid for nonconvex smooth projective targets using virtual classes.

Lemma 2.1; (4)–(5) · p. 5 ·

Correlator relations [KM] · (4a) · print p. 388 / PDF p. 4; Lemma 1.4, (12) · print p. 391 / PDF p. 7 ·

Rational curves on hypersurfaces [PA] · §1.2 · print p. 311 / PDF p. 6

2. Finite shifts and factorial normalization

Set $w_{\beta,k} = (d_\beta + k)!v_{\beta,k}$. Then $(A_j(E)w)_{\beta,k} = \beta_j w_{\beta,k} / (d_\beta + k)$. Combining these identities with the coefficients of $-K_X$ and inverting the grading-raising classical term by a finite nilpotent series gives $\|w_\beta\| \leq B^{d_\beta+1}$. The positive shifts here form a finite set and decrease anticanonical degree. The shifts arising from the polynomial relation below need not have this monotonicity.

§4.1; (6)–(7) · pp. 11–12

3. A free direction avoiding a relation

Suppose every joint diagonal tuple over $L = \overline{\mathbb{C}(q_1, \dots, q_r)}$ is algebraically dependent. Multiplying finitely many relations, then using simultaneous triangularization and homogeneity (3), gives a nonzero homogeneous polynomial Q with $Q(A_1, \dots, A_r) = 0$. Lemma 3.3 supplies a free degree η with $Q(\eta) \neq 0$. That lemma uses rational connectedness of Fano manifolds, Tian–Zong's generation of $\mathbf{CH}_1$ by rational curves, and comb smoothing. Taking handle $\mathbb{P}^1$ and a twist of degree -1 gives $H^1(f^*T_X(-1)) = 0$ after smoothing, hence freeness. Every rational curve degree is a difference of free degrees; closure under addition then gives Zariski density.

Lemma 3.3 · pp. 10–11; Proposition 4.1 (proof) · p. 12 ·

One-cycles [TZ] · Theorem 1.3 · p. 2; Proposition 2.4 · p. 4 ·

Fano rational connectedness [CA] · Corollary 3.2 · print p. 543 / PDF p. 6 ·

Rational curves [DE] · Definition 2.20 · print p. 29 / PDF p. 30; Theorem 2.49 · print p. 43 / PDF p. 44

4. Linearly many iterations force incompatible decay

Polynomial differencing with $p = \beta/d_\beta$ held fixed gives $\|w_\beta\| \leq (C/d_\beta) \max_{\delta \in U} \|w_{\beta-\delta}\|$ near η/d_η . For $\beta = s\eta$, choose $\epsilon > 0$ so that the degree and direction remain controlled through $\lfloor \epsilon d_\beta \rfloor$ finite shifts. Using (7) after these iterations yields

$$\|w_\beta\| \leq (2C/d_\beta)^{\lfloor \epsilon d_\beta \rfloor} B^{2d_\beta+1}.$$

On the other hand, $s\eta$ remains free, and Proposition 3.1 gives

$$(w_\beta, 1) \geq \frac{(d_\beta + n)!}{(ad_\beta)^{d_\beta-1}} \geq ad_\beta^{n+1} (ae)^{-d_\beta}.$$

The logarithm of the upper bound is $-\epsilon d_\beta \log d_\beta + O(d_\beta)$, whereas the lower bound is $-O(d_\beta)$. Their incompatibility proves the algebraic independence asserted in Proposition 4.1.

Proposition 4.1; (8)–(11) · pp. 11–13

5. Proof of Proposition 5.3 — trace and inequality

Proof target : Proposition 5.3; (15) · p. 15

Proposition 5.3 — Independent curve correspondences

Suppose the quantum divisor operators $A_1(q), \dots, A_r(q)$ have a joint eigenvalue tuple algebraically independent over $\mathbb{C}$. Let S_γ be the two-point correspondence from the preceding section and put $d_\gamma = -K_X \cdot \gamma$. The manuscript asserts that there exist linearly independent numerical degrees $\gamma_1, \dots, \gamma_r$ of positive genus-zero stable maps such that

$$S_{\gamma_1} \cdots S_{\gamma_r} \neq 0, \quad \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n.$$

Consequently $r(\iota - 1) \leq n$. Proposition 4.1 supplies the independent tuple required here.

Proposition 5.3; (15) · p. 15

Independence of eigenvalues alone does not yet exhibit a nonzero product of correspondences in the fixed cohomology basis. Lemma 5.1 handles the derivatives of a basis depending on the variables.

1 Select an independent tuple by interpolation

$A_0 = p(A_1, \dots, A_r), \quad p(\lambda) = 1, \quad p(\mu) = 0 \ (\mu \neq \lambda)$
 Use simultaneous triangularization; no semisimplicity is assumed.

Pair permutations to cancel commutators from the differentiated basis.

$$[A_j, B_\ell] = [A_\ell, B_j]$$

[Lemmas 5.1–5.2 · pp. 13–15](#)

2 A conjugation-invariant alternating trace

$$\Theta = \sum_{\pi \in \mathfrak{S}_r} \text{sgn}(\pi) \text{Tr}(A_0 dA_{\pi(1)} \cdots dA_{\pi(r)})$$

$$\Theta = r! m_\lambda d\lambda_1 \wedge \cdots \wedge d\lambda_r \neq 0$$

Choose a term with both wedge and trace nonzero.

$$\omega_\gamma = \sum_\ell \gamma_\ell dq_\ell / q_\ell$$

[Proposition 5.3; \(15\) · p. 15](#)

3 Extract a nonzero term in the fixed basis

$$\omega_{\gamma_1} \wedge \cdots \wedge \omega_{\gamma_r} \neq 0, \quad S_{\gamma_1} \cdots S_{\gamma_r} \neq 0$$

The degree vectors $\gamma_1, \dots, \gamma_r$ are linearly independent.

A nonzero product cannot lower degree beyond $H_0, \dots, H_n$.

[§2 · pp. 3–4](#)

[Proposition 5.3 · p. 15](#)

4 Count the total codimension drop

$$S_\gamma(H_k) \subset H_{k+1-d_\gamma}, \quad r(\iota - 1) \leq \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n$$

An invariant alternating trace detects independent curve degrees and yields the grading inequality

[Open figure ↗](#) [TeX source ↓](#)

1. Alternation removes derivatives of the basis

When A_0 commutes with every A_j , the manuscript proves simultaneous conjugation invariance of

$$\Theta(A_0; A_1, \dots, A_r) = \sum_{\pi \in \mathfrak{S}_r} \text{sgn}(\pi) \text{Tr}(A_0 dA_{\pi(1)} \cdots dA_{\pi(r)}).$$

A change of basis $G(q)$ introduces $K = G^{-1}dG$. Setting $B_j(t) = dA_j + t[K, A_j]$ gives $[A_j, B_\ell(t)] = [A_\ell, B_j(t)]$. Pairing permutations with exchanged labels cancels the commutator terms in the differentiated trace. The proof does not differentiate A_0 or require A_0 to commute with dA_j .

Lemma 5.1; (12)–(14) · pp. 13–14

2. Selecting one joint tuple

Choose an interpolation polynomial p taking value one at the independent tuple λ and zero at the other distinct joint tuples, and set $A_0 = p(A_1, \dots, A_r)$. In a common upper triangular basis,

$$\Theta = r! m_\lambda d\lambda_1 \wedge \dots \wedge d\lambda_r \neq 0.$$

The matrix A_0 need not be an idempotent projector on a generalized eigenspace: selecting the diagonal entries suffices. This is another point at which the argument avoids semisimplicity.

Lemma 5.2 · pp. 14–15

3. Returning to the fixed basis and counting degrees

In the fixed basis, $dA_j = \sum_{\gamma > 0} q^\gamma \gamma_j S_\gamma \omega_\gamma$, where $\omega_\gamma = \sum_\ell \gamma_\ell dq_\ell / q_\ell$. A nonzero summand in the expansion of Θ has both $\omega_{\gamma_1} \wedge \dots \wedge \omega_{\gamma_r} \neq 0$ and $\text{Tr}(A_0 S_{\gamma_1} \dots S_{\gamma_r}) \neq 0$. The first gives linear independence of degree vectors; the second gives a nonzero ordered operator product. Each correspondence lowers codimension by $d_{\gamma_j} - 1$. Since the available degrees are $H_0, \dots, H_n$,

$$r(\iota - 1) \leq \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n.$$

This is Proposition 5.3 and proves the inequality in Theorem 1.1.

Proposition 5.3; (15) · p. 15

6. Proof of Proposition 6.4 — classification at equality

Proof target : Proposition 6.4 · p. 17

Proposition 6.4 — Rigidity at equality

Suppose $\iota \geq 2$, $r(\iota - 1) = n$, and the independent nonzero product in (15) of Proposition 5.3 exists: there are linearly independent positive stable-map degrees $\gamma_1, \dots, \gamma_r$ with

$$S_{\gamma_1} \cdots S_{\gamma_r} \neq 0, \quad \sum_{j=1}^r (d_{\gamma_j} - 1) \leq n.$$

Then the manuscript asserts

$$X \simeq (\mathbb{P}^{\iota-1})^r.$$

For the main theorem, Proposition 5.3 provides the existence of this nonzero product.

Proposition 6.4 · p. 17

Assume $r(\iota - 1) = n$. Both inequalities above must be equalities, forcing every selected degree to satisfy $d_{\gamma_j} = \iota$. Only now does the argument pass from a nonzero correspondence product to actual families of minimal rational curves.

1 Equality makes every selected degree minimal

$$r(\iota - 1) = n, \quad d_{\gamma_j} = \iota, \quad S_{\gamma_1} \cdots S_{\gamma_r} \neq 0$$

The supports give a nonempty incidence chain, ordered by composition.

Use evaluation supports and stable limits to choose full components.

[Lemma 6.3 · pp. 16–17](#)

2 Take full irreducible components and obtain properness

$$V^j \subset \text{RatCurves}^n(X), \quad -K_X \cdot V^j = \iota$$

Minimal degree excludes splitting and multiple covers; classes are independent.

Check the corrected hypothesis: proper irreducible components.

[Mukai chain bound \[BCDD\] · Theorem 5.2 · print p. 623 / PDF p. 23](#)

[Mukai chain bound \[ERR\] · Erratum · p. 1](#)

[Theorem 6.1 · p. 16](#)

[Proposition 6.4 · p. 17](#)

3 Endpoint dimension and cyclic rotation

$$\dim \text{Locus}(V^1, \dots, V^r)_x \geq r(\iota - 1) = n$$

The last family covers; rotate the chain to make every family covering.

Use unsplit, covering, independent families of degrees $n_j + 1$.

[Product characterization \[OC\] · Theorem 1.1 · print p. 271 / PDF p. 2](#)

[Proposition 6.4 · p. 17](#)

4 Apply the product characterization

$$n_j = \iota - 1 > 0, \quad \sum_j n_j = n, \quad X \simeq (\mathbb{P}^{\iota-1})^r$$

Minimal nonzero correspondences supply proper curve families, coveringness, and the product characterization

[Open figure ↗](#) [TeX source ↓](#)

1. From supports of correspondences to a nonempty chain

The two-evaluation image $Z_\gamma \subset X \times X$ contains the support of the correspondence inducing S_γ . If the incidence fiber product matching consecutive evaluations were empty, the composition would vanish. A nonzero product therefore supplies a chain. Order the curves according to composition, reversing and relabelling the degree list if necessary. Total degree ι forces each stable map to have exactly one nonconstant component, birational onto its image. Take the **full irreducible components** of $\text{RatCurves}^n(X)$ containing these curves. After forgetting the markings, a stable limit has neither splitting, multiple covers, nor contracted trees. These components are consequently proper.

[Lemma 6.3 · pp. 16–17](#)

2. The ordered-locus bound makes every family covering

Theorem 5.2 of the Mukai chain bound gives dimension at least $\sum_j (-K_X \cdot V^j - 1)$ for a nonempty ordered endpoint locus of numerically independent proper irreducible components. The erratum requires irreducible components of $\mathbf{RatCurves}^n(X)$; arbitrary proper subfamilies do not suffice. Lemma 6.3 provides precisely this condition.

At equality the lower bound is n , so the last family V^r is covering. Properness makes its evaluation image closed, hence all of X . Prepending a member of V^r to the original chain and dropping the last member rotates the order. Repeating this operation puts each V^j last in a nonempty chain. The same dimension bound makes every family covering.

[Theorem 6.1 · p. 16](#); [Proposition 6.4 · p. 17](#) ·

[Mukai chain bound \[BCDD\] · Theorem 5.2 · print p. 623 / PDF p. 23](#) ·

[Mukai chain bound \[ERR\] · Erratum · p. 1](#)

3. Checking Occhetta's hypotheses

The families are unsplit, covering, numerically independent, and all have anticanonical degree ι . With $n_j = \iota - 1 > 0$, one has $\sum_j n_j = n$, so Occhetta's Theorem 1.1 gives $X \simeq (\mathbb{P}^{\iota-1})^r$. Conversely, computing the dimension, Picard number, and pseudoinde x of this product proves equality.

[Theorem 6.2 · p. 16](#); [Proposition 6.4; converse · p. 17](#) ·

[Product characterization \[OC\] · Theorem 1.1 · print p. 271 / PDF p. 2](#)

7. Which papers supply which steps

These inputs and alternative constructions are external to catalogue 063. The Mustață–Mustață construction is an alternative to the contraction explicitly constructed in the manuscript. Their statements have been compared with the manuscript's applications; complete proofs of the external results have not been independently verified.

Source and result

Correlator relations [KM] · (4a) · print p. 388 / PDF p. 4; Lemma 1.4, (12) · print p. 391 / PDF p. 7; Rational curves on hypersurfaces [PA] · §1.2 · print p. 311 / PDF p. 6

Used at

Lemma 2.1, p. 5

Role and required hypotheses

Genus-zero recursion, divisor and string identities; smooth projective target, positive-degree forgetful morphism, $D_j \cup \mathbf{pt} = 0$

Checking scope

Input formulas, the nonconvex qualification, and application checked.

Source and result

Intermediate moduli [MM] · Definition 1.2; Proposition 1.3 · pp. 5–6; Lemma 3.3 (proof) · pp. 17–18; Chow ring [MMC] · Proposition 1.7 · p. 5

Used at

§§3.2–3.3, pp. 7–8

Role and required hypotheses

Alternative contraction away from the marking, weighted coefficients, cotangent class; projective-space target in characteristic zero

Checking scope

Alternative contraction and cotangent class checked; virtual positivity is argued in §3.4.

Source and result

Fano rational connectedness [CA] · Corollary 3.2 · print p. 543 / PDF p. 6; Rational curves [DE] · Definition 2.20 · print p. 29 / PDF p. 30; Theorem 2.49 · print p. 43 / PDF p. 44

Used at

Lemma 3.3, p. 10

Role and required hypotheses

Smooth complex projective Fano implies rational chain connectedness, then separable rational connectedness

Checking scope

Checked while retaining the distinction from Campana’s older terminology.

Source and result

One-cycles [TZ] · Theorem 1.3 · p. 2; Proposition 2.4 · p. 4

Used at

Lemma 3.3, pp. 10–11

Role and required hypotheses

Rational-curve generation and comb smoothing; smooth proper SRC target, handle $\mathbb{P}^1$, twist -1

Checking scope

Generation by rational curves and passage to numerical degrees checked.

Source and result

Mukai chain bound [BCDD] · Theorem 5.2 · print p. 623 / PDF p. 23 and Mukai chain bound [ERR] · Erratum · p. 1

Used at

Theorem 6.1 and Proposition 6.4, pp. 16–17

Role and required hypotheses

Proper **irreducible components**, independent numerical classes, nonempty ordered chain

Checking scope

Original statement, erratum, and connection from Lemma 6.3 checked.

Source and result

Product characterization [OC] · Theorem 1.1 · print p. 271 / PDF p. 2

Used at

Theorem 6.2 and Proposition 6.4, pp. 16–17

Role and required hypotheses

Independent unsplit covering families, degrees $n_j + 1$, $\sum n_j = n$

Checking scope

Statement and application after cyclic rotation checked.

Givental's quantum differential equations and Khalkhali's generalized trace are background references cited by the manuscript. Lemmas 2.1 and 5.1 derive the particular identities needed here; the background references have not been turned into additional direct dependency edges. Relations with the existing catalogues 033 and 034 have not been exhaustively investigated.

8. Return to the sources

- [Proposition 3.1](#) · p. 6; [Proposition 4.1](#) · p. 11 — The descendant lower bound and the independence conclusion.
- [§4](#) · pp. 11–13 — Iteration of the recurrence and comparison with the lower bound.
- [Lemmas 5.1–5.2](#) · pp. 13–15 — Cancellation of basis derivatives and selection of a joint eigenvalue tuple.
- [Proposition 5.3; \(15\)](#) · p. 15 — The nonzero trace term yields the inequality.
- [Theorem 6.2](#) · p. 16; [Proposition 6.4; converse](#) · p. 17 — Minimal rational-curve families classify the equality case.

The proof passages in §§2–6 were read, and the external statements listed above were compared with their uses. Particular attention was paid to the unstable degree-zero term, virtual pushforward, factorial normalization, derivatives of the basis, and the corrected family hypotheses. This is an overview of the manuscript's argument, not an independent certification of its correctness.

Outstanding verification includes the foundations of virtual classes and associativity, all technical compatibility of the family contraction and stack descent in §3.2, a complete independent check of the commutator calculation in Lemma 5.1, and the full proofs of external inputs. Complete dependencies with existing articles remain uninvestigated. These limits are distinct from the statement-and-application checks recorded above.

[§§2–6](#) · pp. 3–17 · [Detailed source and checking record](#)